

Zhang Yangruo

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Address: Zhuhai, Guangdong, China

EDUCATION

Beijing Normal-Hong Kong Baptist University 09/2023 – Present
Bachelor of Science (Honours) in Applied Mathematics GPA: 3.41/4.0
Minor in Computer Science and Technology Expected Graduation: Jun. 2027
Current Degree Classification: First Class Honours
Main Coursework: Probability Theory, Mathematical Statistics, Applied Stochastic Process, Optimization, Numerical Analysis, Mathematical Modelling, Machine Learning, Data Structures and Algorithms, Database Management Systems, Graph Theory, Discrete Mathematics

RESEARCH EXPERIENCE

Decoding Cellular State Trajectories in the Tumor Microenvironment Sep. 2025 – Present
First Author; Final-Year Project under the supervision of Dr. Jiaxing Chen

Manuscript in preparation; target submission to a computational biology journal in Jul. 2026

- Developing a statistical and computational framework for reconstructing heterogeneous multi-root cellular trajectories in the tumor microenvironment from single-cell and spatial transcriptomic data.
- Designed a multi-stage methodology integrating root identification, multi-root tree construction, conditional mutual information, causal structure analysis, and local progression modelling to characterize heterogeneous cell-state progression trajectories.
- Extended trajectory analysis through causal vector fields, global potential fields, and transfer entropy to quantify progression direction, convergence behaviour, and interactions between competing cellular trajectories.
- Evaluated the framework across multiple single-cell and spatial transcriptomic datasets, including Xenium datasets from breast, lung, and melanoma studies, using synthetic-data validation, benchmarking, sensitivity analysis, and bootstrap stability assessment.

ADAPT: Forecast-Guided Adaptive Diagnostic Triage for Dengue Jan. 2026 – Apr. 2026
First Author; Manuscript submitted to Computer Methods and Programs in Biomedicine

- Developed ADAPT, a forecast-guided decision-support framework linking weekly dengue RT-PCR-confirmed positivity forecasting with cost-sensitive PCR triage.
- Processed 78,068 dual-tested Brazilian SINAN dengue surveillance records from 2021–2025 and constructed a reproducible weekly positivity time series.
- Implemented a bounded-output attention-guided LSTM in PyTorch, achieving held-out test MSE of 0.091, MAE of 0.227, and Pearson correlation of 0.719.
- Derived a closed-form analytical decision threshold and used Monte Carlo simulation to quantify policy robustness under uncertainty in diagnostic sensitivity and specificity.
- In an 18-week held-out evaluation, ADAPT improved PPV from 0.907 to 0.917 and reduced expected false positives from 12.99 to 11.43 relative to static NS1 screening.

TME Hypergraph-based Perturbation Response Prediction Sep. 2025 – Jan. 2026
Third Author; Manuscript submitted to Bioinformatics

- Addressed limitations of existing single-cell perturbation-response models in capturing higher-order spatial organization and local niche signals within the tumor microenvironment.
- Contributed to a semantic hypergraph-based framework that encodes biologically motivated microenvironmental motifs and extends the chemCPA baseline for rigorous benchmarking.
- Synthesized benchmark evaluation results across multiple datasets and contributed to model evaluation, result analysis, and manuscript preparation.

INTERNSHIP EXPERIENCE

Chia Tai Tianqing Pharmaceutical Group Co., Ltd. Jiangsu, China
Biopharmaceutical Development Research Center – Biologics R&D Assistant Jun. 2026 – Present

- Conducted literature review and biomedical data collection and organization to support biologics research and development projects.

- Performed data cleaning, descriptive statistical analysis, and data visualization for biomedical research datasets.
- Supported research documentation and data preparation workflows, improving data organization, consistency, and readiness for subsequent analysis.

Machang Tianyuan Surveying and Mapping Co., Ltd.

Anhui, China

Remote Sensing Data Analyst

Jan. 2026 – Mar. 2026

- Validated satellite-based inversion models for vegetation-cover and water-quality indicators using regression analysis, hypothesis testing, and statistical reliability assessment.
- Applied Monte Carlo simulations to quantify uncertainty propagation in remote-sensing-derived environmental parameter estimates.
- Conducted time-series analysis of seasonal vegetation-index patterns to support environmental monitoring and model validation.

Jiangsu Hoperun Software Co., Ltd.

Jiangsu, China

Huawei Enterprise Department – Assistant Data Annotator

Jun. 2025 – Aug. 2025

- Processed and annotated enterprise datasets for Huawei-related machine learning workflows, supporting the construction of high-quality ground-truth data.
- Performed data quality assurance by identifying anomalies, resolving edge cases, and improving annotation consistency across unstructured datasets.
- Supported dataset preprocessing and quality-control workflows to improve data reliability for downstream machine learning tasks.

HONORS AND ACADEMIC ACTIVITIES

China Undergraduate Mathematical Contest in Modeling – Provincial Second Prize	Oct. 2025
China Undergraduate Mathematical Contest in Modeling – Provincial Third Prize	Oct. 2024
BNBU Undergraduate Competition Awards	Apr. 2026
Imperial College London – Certificate in Statistics & Data Analysis in Public Health	Jan. 2026
Johns Hopkins University – Biostatistics in Public Health Specialization	Apr. 2026 – May. 2026
Leader Path Programme of BNBU – Gold Award	May. 2026
Leader Path Programme of BNBU – Silver Award	May. 2025
The 14th National Forum on Mathematical Culture – Conference Attendee	Jun. 2025 – Jul. 2025

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

• Executive Committee Member , BNBU Student Union	Sept. 2023 – Jun. 2025
• Student Administrative Assistant , BNBU Admission Office	May. 2024 – Jul. 2024
• Peer Mentor , BNBU Peer Mentor Program	Sept. 2024 – Jun. 2025
• Chairperson of Electoral Commission , BNBU Student Union	Sept. 2025 – Jun. 2026
• Head of Organization Department , BNBU CCYL Committee	Sept. 2025 – Jun. 2026
• Resident Tutor , BNBU Student Affairs Office	Sept. 2025 – Jun. 2026
• Participant , Guangdong province “Qingma Project” Youth Leadership Training Program	Jul. 2026

SKILLS

- **Programming & Tools:** Python, PyTorch, MATLAB, C/C++, MySQL, Linux, Matplotlib, Seaborn
- **Statistical & Computational Methods:** Bayesian Inference, Causal Inference, Conditional Mutual Information, Monte Carlo Simulation, Time-Series Forecasting, Transfer Entropy, Bootstrap Stability Analysis, Uncertainty Quantification
- **Bioinformatics & Machine Learning:** Single-Cell Analysis, Spatial Transcriptomics, Trajectory Inference, Tumor Microenvironment Modelling, LSTM, Attention-based Models, Graph Neural Networks, Statistical Machine Learning
- **Languages:** Mandarin Chinese (Native), English (IELTS Academic 6.5)